

🌟 1. Simple Meaning

💡 *Attention* helps a model **focus on the most important parts** of the input — just like your brain focuses on key words in a sentence.

📖 Example:

“The cat sat on the mat because **it** was tired.”
When predicting “it”, the model **attends** to “cat”, not “mat”.

⚡ 2. Why We Need Attention

- 🌀 Old models (RNNs, LSTMs) → read words one by one
- ❌ Forget long-range information
- ✅ Attention → allows model to “look at” **all words at once** and decide what matters most

🧩 3. Main Idea

Each word creates three vectors:

- 🔹 **Q (Query)** → What I want to know
- 🔹 **K (Key)** → What each word offers
- 🔹 **V (Value)** → Actual meaning or info of each word

The model finds how much each word’s **Key** matches the **Query**, and uses those scores to mix the **Values** = focused output ✨

📊 4. The Core Formula

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V$$

- 💬 Meaning:
1. Compute similarity → QK^T
 2. Scale → divide by $\sqrt{d_k}$
 3. Normalize → softmax (make probabilities)
 4. Weight → multiply by V to get focused info

📋 5. Step-by-Step Flow

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Input Words → [Q, K, V] → Dot Product → Softmax → Weighted Sum → Output

🖼️ Example Visualization:

Word	Attention Weight (for “it”)
The	0.05
cat	🟢 0.70
sat	0.10
on	0.05
mat	0.10

✅ So the model focuses **70%** on “cat” → correct context!

6. Types of Attention

Type	Description	Example Use
Additive (Bahdanau)	Uses small NN for attention score	Early seq2seq models
Dot-Product (Luong)	Simpler, uses dot product	Translation models
Scaled Dot-Product	Divides by $\sqrt{d_k}$ for stability	Transformers
Self-Attention	Each word attends to <i>all others</i> in the same sentence	Transformer Encoder
Cross-Attention	Decoder attends to encoder outputs	Translation
Multi-Head Attention	Many attention layers in parallel	Transformer models

7. Self-Attention (Most Important)

Every word learns about every other word.
So the sentence's meaning becomes **context-aware**.

Example:

"Bank of the river" vs "Bank of America"
Self-Attention helps the model understand *which* meaning of "bank" to use 🗣️ 🏦

8. Real-Life Example

👤 You're in a classroom.

You listen to everything, but focus mainly on the teacher's key points.

That's **attention**!

9. Benefits

- ✓ Understands long sentences easily
- ✓ Learns global relationships
- ✓ Works in parallel (fast!)
- ✓ Forms the base of **Transformers, BERT, GPT**

11. In One Line

💡 **Attention = Learn to focus.**
Instead of remembering everything, focus only on what matters.

12. Connection to Transformer

- Attention was first used inside **RNNs (2014)**
- Later (2017), **Transformers** said —
👉 "Let's remove RNNs completely and use *only attention*."

📖 So:

Attention = building block
Transformer = building made from attention

🎯 Example: How Attention Works

Let’s take a small English-to-Telugu translation example 🗣️

💬 Sentence:

Input (English): “I love apples”

Output (Telugu): “నాకు ఆపిల్స్ ఇష్టం”

When predicting each Telugu word, the model decides **which English word to focus on most**.

🧠 Step-by-Step Process

🔮 Step 1: Inputs → Vectors

Each English word gets converted into numerical vectors:

Word	Vector (simplified)
I	[1, 0, 0]
love	[0, 1, 0]
apples	[0, 0, 1]

⚙️ Step 2: Create Q, K, and V Matrices

From each word, the model creates:

- **Q (Query)** → What am I focusing on?
- **K (Key)** → What information do I have?
- **V (Value)** → The actual content/info

(They are learned by multiplying each word vector with weight matrices.)

📊 Step 3: Compute Attention Scores

For each **Query**, the model computes **dot products** with all **Keys**.

Example when predicting “ఇష్టం (like)” →

the model compares “like” (Query) with each English word’s Key:

English Word	Similarity Score (Q·K)	Meaning
I	0.8	somewhat related
love	1.0	very related ❤️
apples	0.3	not very related

📐 Step 4: Apply Softmax

Convert scores to probabilities (attention weights):

Word	Raw Score	Softmax Weight
I	0.8	0.35
love	1.0	🟢 0.50
apples	0.3	0.15

✅ So “love” gets the **highest attention weight (0.50)**.

🧠 Step 5: Weighted Sum (Combine Values)

The model multiplies each **Value (V)** by its weight and adds them up → that gives a **context vector** focused on the right info — “love”.

⚡ Step 6: Output

Now the model uses that **context vector** to generate the next Telugu word →

🗨 Predicts: “ఇష్టం (love)”

So the model successfully focused on the **right English word** when generating each Telugu word. 🎉

💡 Summary Visualization

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Input: I → love → apples

👉 👉 👉

(Attention Weights)

👉 👉 👉

Output: నాకు ఆపిల్స్ ఇష్టం

When predicting “ఇష్టం”, the arrow from “love” is strongest — that’s **attention** in action 🔥



📖 ATTENTION ARCHITECTURE — Full Concept + Visual Notes

🎯 1. Core Idea

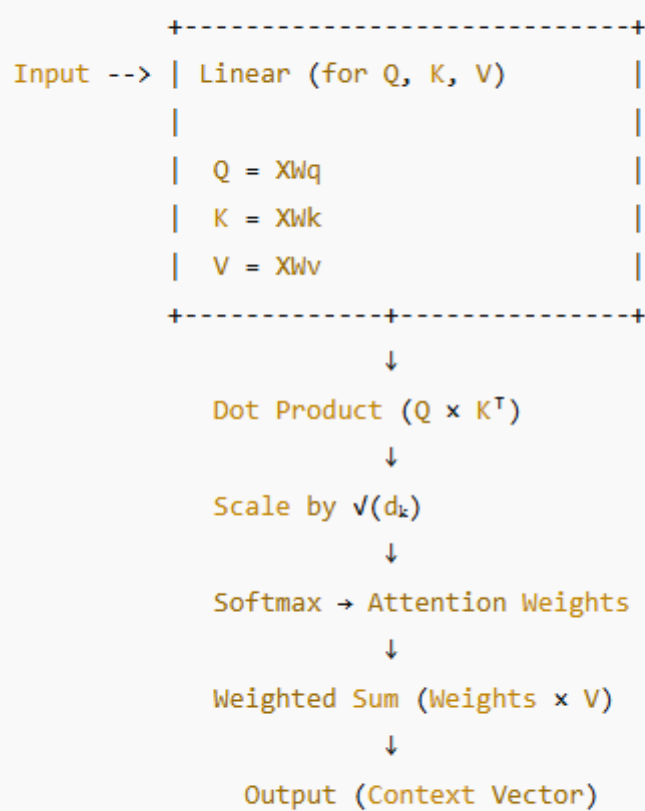
Attention = A small *sub-network* that decides “which input parts to focus on” when producing an output.

It takes:

- 🔹 Queries (Q)
 - 🔹 Keys (K)
 - 🔹 Values (V)
- and produces a **context vector** (weighted information).

⚙️ 2. Basic Attention Block Architecture

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- ✓ Output = “focused” representation of input
- ✓ Used everywhere inside Transformer

🔹 3. The Mathematical Architecture

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V$$

- $Q = XW_q$
- $K = XW_k$
- $V = XW_v$

Where:

- $X \rightarrow$ input embeddings
- $W_q, W_k, W_v \rightarrow$ learnable weight matrices

💡 4. Self-Attention Architecture

In **self-attention**,

Q, K, and V **come from the same input sequence**.

```
sql

Sentence: [I, love, apples]

Each word → Embedding → Q, K, V
↓
Compute attention across all words
↓
Each word's output now knows about others
```

So each token gets **context-aware meaning**.

Example:

"bank" in "river bank" vs "bank account"
→ meaning changes based on context captured by self-attention 🗣️ 💰

🔥 5. Multi-Head Attention Architecture

One attention head learns one type of relationship —
but we want multiple relationships (syntax, semantics, etc.).
So Transformers use **Multi-Head Attention (MHA)**.

🧩 Multi-Head Attention Block:

```
css

Input
↓
Linear layers → Create multiple (Q,K,V) sets
↓
Attention Head 1 → captures relation type A
Attention Head 2 → captures relation type B
...
Attention Head h → captures relation type H
↓
Concatenate all head outputs
↓
Linear projection → Final Output
```

Formula:

$$\text{MHA}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W_o$$

where each

$$\text{head}_i = \text{Attention}(QW_q^i, KW_k^i, VW_v^i)$$

- ✅ Each head learns different information
- ✅ Combined = richer context understanding

💬 9. Real Example

For the sentence:

"I love apples" 🍏

Each word learns connections like:

- "I" → attends to "love" (subject–verb relation)
- "love" → attends to "I" and "apples" (verb–object)
- "apples" → attends to "love" (object–action)

All these dependencies are captured **automatically** ✨

🧩 6. Add & Norm Layers (Post-Attention)

After attention, the Transformer adds:

1. **Residual Connection:** adds input back to output
2. **Layer Normalization:** stabilizes learning

```
ini

Output = LayerNorm(Input + AttentionOutput)
```

This keeps gradients stable & improves training 🔄

Simple Attention Example

 **Sentence:**

"I love AI"

We want to see **self-attention** for the word "love".

Step 1 : Represent Words as Vectors

For simplicity, assume each word is a 2D vector:

Word	Vector
I	[1, 0]
love	[0, 1]
AI	[1, 1]

Step 2 : Define Q, K, V Matrices

Let's assume **identity matrices** (for simplicity):

$$Q = K = V = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$$

So each word's vector itself acts as Q, K, and V.

Step 3 : Compute Attention Scores

$$\text{Score} = Q \cdot K^T$$

We compute **dot product** between "love" (Query) and all Keys:

- Query "love" = [0,1]
- Keys = [[1,0], [0,1], [1,1]]

Dot products:

- "I": [0,1]·[1,0] = 0
- "love": [0,1]·[0,1] = 1 
- "AI": [0,1]·[1,1] = 1

So **raw scores** = [0, 1, 1]

Step 4 : Apply Softmax

$$\text{Softmax}([0, 1, 1]) = \left[\frac{e^0}{e^0 + e^1 + e^1}, \frac{e^1}{e^0 + e^1 + e^1}, \frac{e^1}{e^0 + e^1 + e^1} \right]$$

Compute roughly:

- $e^0 = 1$
- $e^1 \approx 2.718$

$$\text{Sum} = 1 + 2.718 + 2.718 \approx 6.436$$

Softmax weights:

- "I" = $1 / 6.436 \approx 0.16$
- "love" = $2.718 / 6.436 \approx 0.42$
- "AI" = $2.718 / 6.436 \approx 0.42$

 So the model pays **most attention** to "love" and "AI"

Step 5 : Compute Weighted Sum of Values

$$\text{Output} = \sum (\text{weight}_i \cdot V_i)$$

Values (V) = [[1,0], [0,1], [1,1]]

Weighted sum:

- $X = 0.16 \cdot [1,0] + 0.42 \cdot [0,1] + 0.42 \cdot [1,1]$
- Compute each component:

$$X_1 = 0.16 \cdot 1 + 0.42 \cdot 0 + 0.42 \cdot 1 = 0.16 + 0 + 0.42 = 0.58$$

$$X_2 = 0.16 \cdot 0 + 0.42 \cdot 1 + 0.42 \cdot 1 = 0 + 0.42 + 0.42 = 0.84$$

 Context vector for "love" = **[0.58, 0.84]**

This is the **attended representation** of "love" considering the whole sentence.

Step 6 : Interpret

- The output vector [0.58, 0.84]
- Has most contribution from "love" (0.42) and "AI" (0.42), less from "I" (0.16)
- Shows **attention mechanism focusing on important words**

1 What is Attention in DL?

- In Deep Learning, **attention** is a mechanism that lets a neural network **focus on the most relevant parts of the input** when producing an output.
- Instead of treating all input equally, the network **weighs important features more**.
 - It's widely used in **NLP, CV, Speech, and Multimodal tasks**.

2 Why Attention?

- Old DL models like **RNNs/LSTMs** had problems:
- ❌ Couldn't remember long sequences easily
 - ❌ Fixed-size context vector → information loss

- Attention solves this:**
- ✅ Allows the model to look at **all parts of input** and choose what's relevant.
 - ✅ Improves translation, summarization, captioning, and even images.

3 Core Components in DL

In Deep Learning, attention is mathematically implemented using:

Component	Description
Query (Q)	What we want to focus on (current state)
Key (K)	Representation of all inputs
Value (V)	Actual information we want to extract
Attention Score	Measures relevance between Q and K
Softmax	Normalizes scores → attention weights
Output	Weighted sum of Values (V) → context vector

4 Types of Attention in DL

Type	DL Use Case	Notes
Additive Attention	Sequence-to-sequence NLP	Uses a small feedforward network
Dot-Product Attention	NLP translation	Efficient and simple
Scaled Dot-Product Attention	Transformers	Divides by $\sqrt{d_k}$ for stability
Self-Attention	Transformers, BERT	Each input element attends to others
Cross-Attention	Transformers (Decoder)	Decoder attends to encoder outputs
Multi-Head Attention	Transformers	Multiple attention heads capture different info

5 Attention Formula in DL

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V$$

- Q, K, V → learned through **linear layers** in deep networks
- Output → context vector that's **fed to next layers**

8 Attention + Deep Learning Flow (Simplified)

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Input Sequence (or Features)

↓

Embedding / Feature Extraction

↓

Compute Q, K, V (Linear Layers)

↓

Attention Scores ($Q \cdot K^T / \sqrt{d_k}$)

↓

Softmax → Attention Weights

↓

Weighted Sum of Values → Context Vector

↓

Feed Forward / Next Layer

Step 1: Input Representation

- Start with a sequence of data (words in NLP, patches in images, features in general).
- Convert inputs into **vectors/embeddings**.

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Input Sequence → Embedding Layer → Vector Representation

Example (NLP):

“I love AI” → [[1,0], [0,1], [1,1]]

Step 2: Create Q, K, V

- Use **linear layers** to transform input embeddings into:
 - **Query (Q)** → what we want to focus on
 - **Key (K)** → all input representations
 - **Value (V)** → actual information to extract

CSS

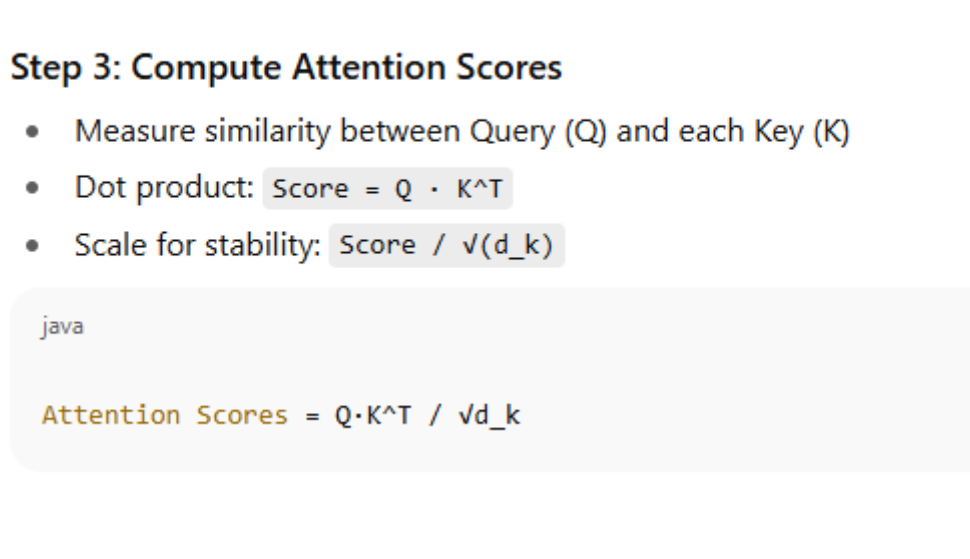
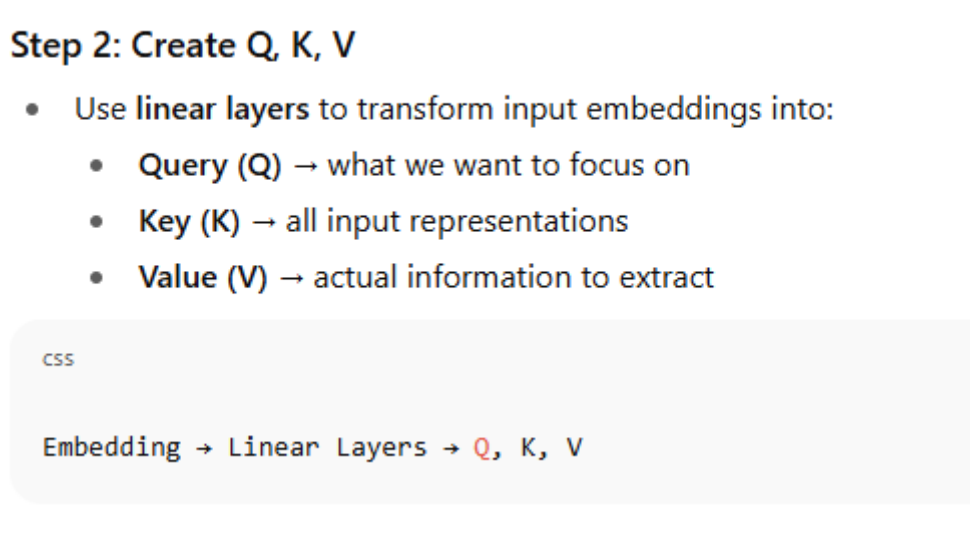
Embedding → Linear Layers → Q, K, V

Step 3: Compute Attention Scores

- Measure similarity between Query (Q) and each Key (K)
- Dot product: `Score = Q · K^T`
- Scale for stability: `Score / √(d_k)`

java

Attention Scores = Q · K^T / √d_k



Step 4: Apply Softmax

- Convert raw scores into **attention weights (probabilities)**
- Higher weight → more focus on that part of input

SCSS

Softmax(Attention Scores) → Attention Weights

Step 5: Weighted Sum

- Multiply attention weights with corresponding Values (V)
- Sum them → **context vector** (focused representation)

java

Context Vector = Σ (Attention Weight × V)

Step 6: Feed Forward

- Pass context vector through a **Feed Forward Network**
- Optionally use **Residual + Layer Normalization**

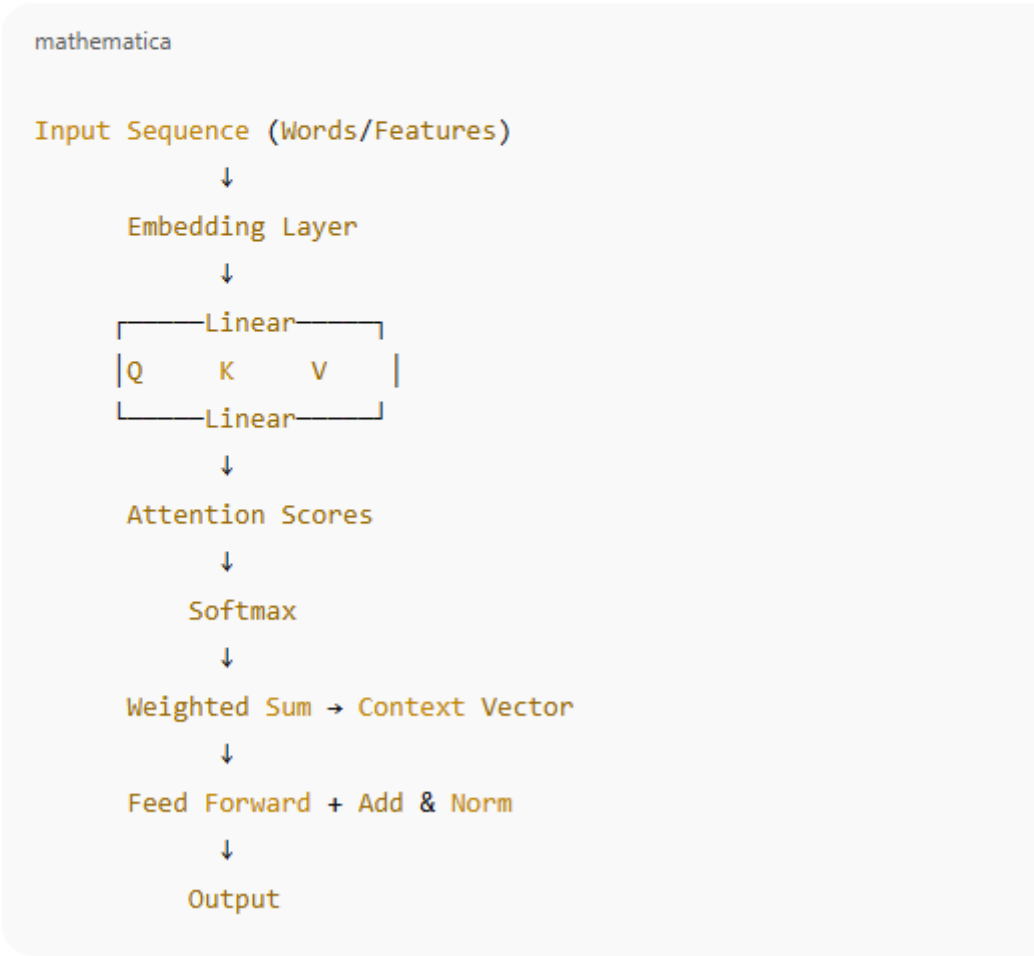
mathematica

Context Vector → Feed Forward → Output Representation

Step 8: Output

- Output = **context-aware representation**
- Can be used for next layer, prediction, translation, or generation

Flow Diagram (Stylish Version)



Step 7: (Optional) Multi-Head Attention

- Repeat the above with multiple attention heads
- Each head captures **different relationships**
- Concatenate heads → Linear projection → Final output

```

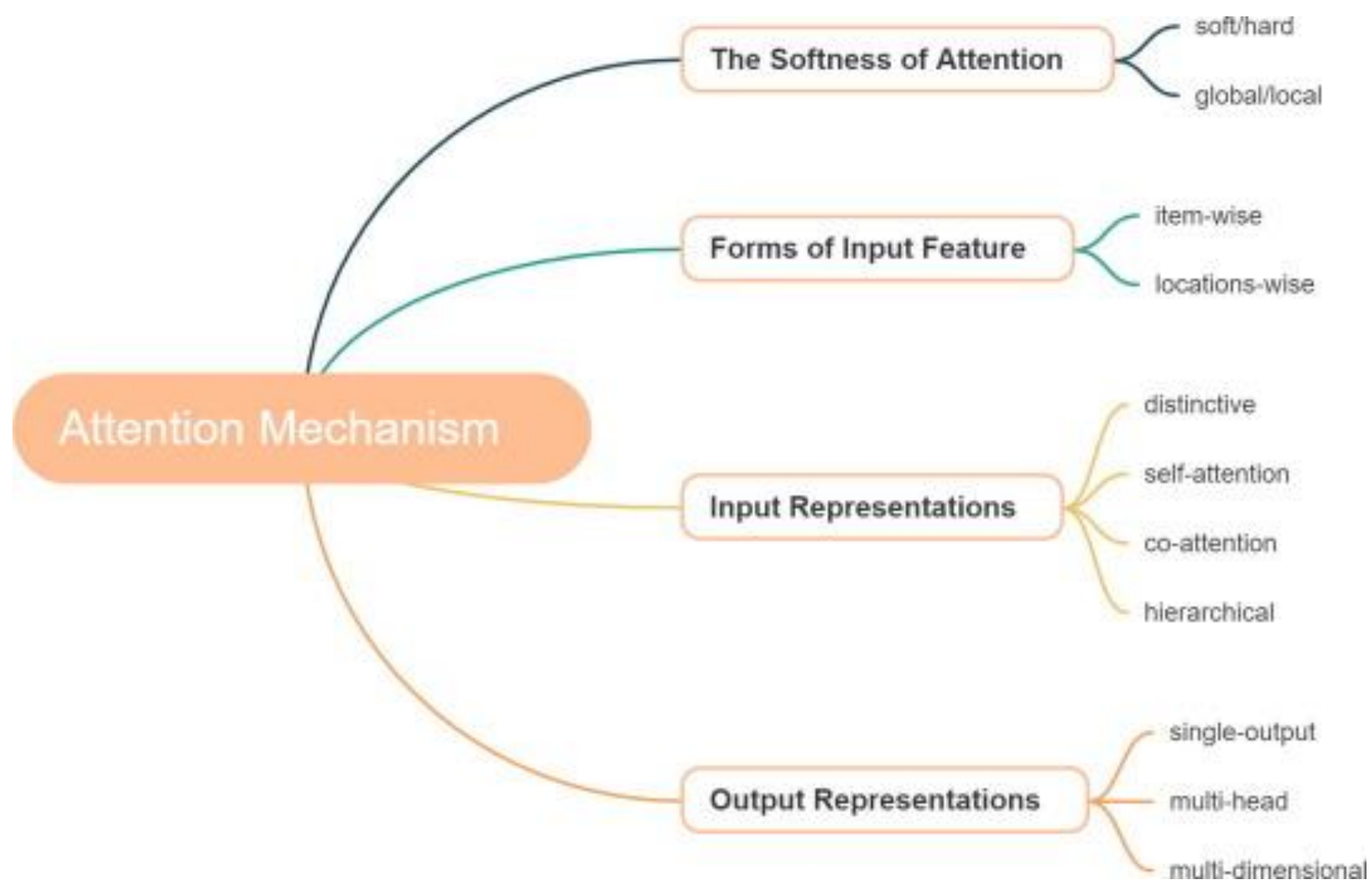
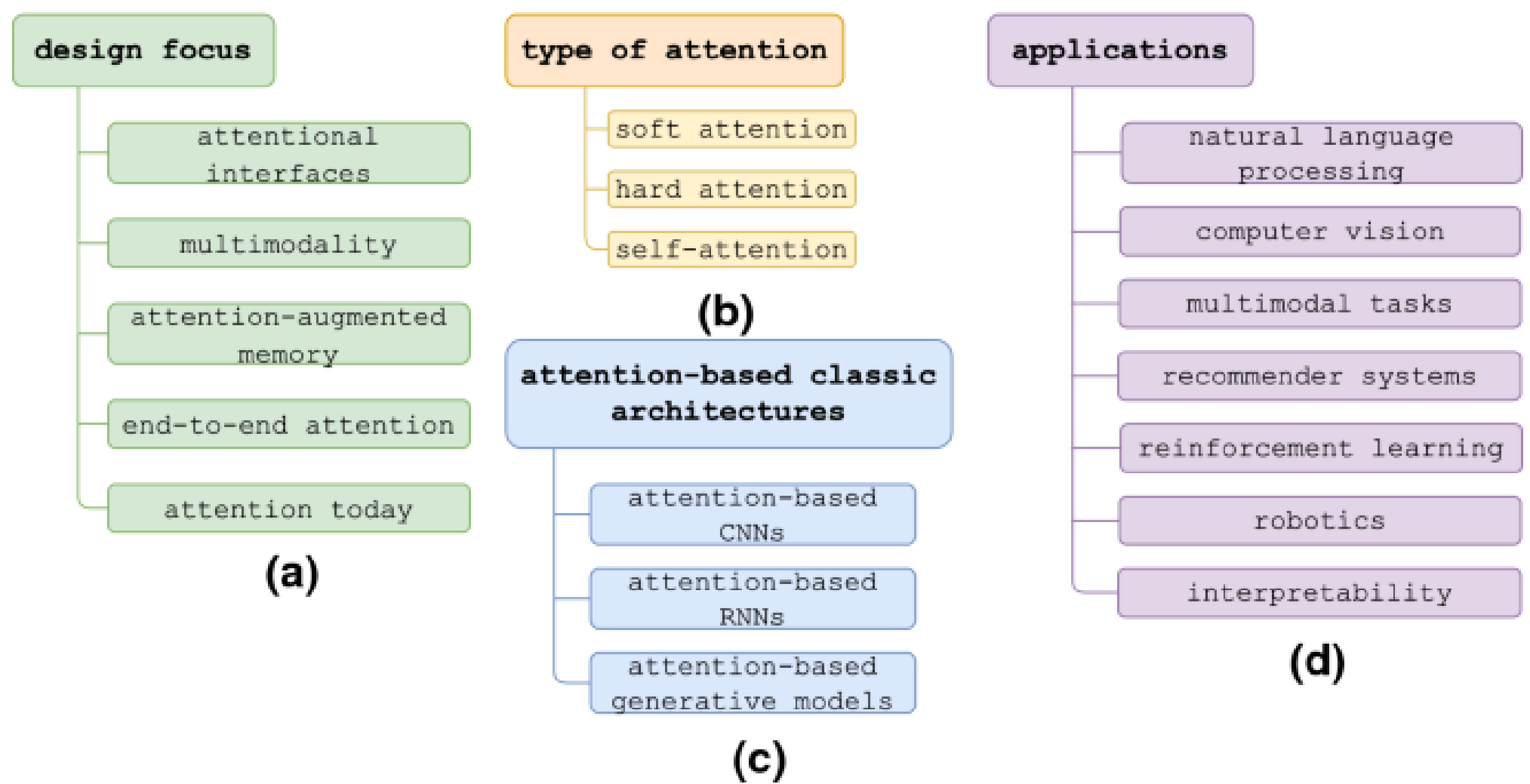
csharp

[Head1, Head2, ..., HeadH] → Concat → Linear → Output

```

◆ Training Flow Diagram (Stylish)





☀ Types of Attention in Deep Learning

Attention mechanisms are mainly classified based on **how queries, keys, and values interact** and **what relationships they capture**.

1 Additive (Bahdanau) Attention

- **Introduced:** 2014 (Bahdanau et al.)
- **How it works:** Computes attention score using a small **feed-forward neural network**.
- **Formula:**

$$score = v^T \tanh(W_q Q + W_k K)$$

- **Use case:** Early **seq2seq machine translation** models.
- **Pros:** Handles variable-length sequences well.

2 Dot-Product (Luong) Attention

- **Introduced:** 2015 (Luong et al.)
- **How it works:** Computes **dot product** between query and key.

$$score = Q \cdot K^T$$

- **Use case:** NLP translation and alignment tasks.
- **Pros:** Simpler and faster than additive attention.

3 Scaled Dot-Product Attention

- **Used in:** Transformers (2017)
- **How it works:** Same as dot-product but scales by $\sqrt{d_k}$ to stabilize gradients:

$$score = \frac{Q \cdot K^T}{\sqrt{d_k}}$$

- **Pros:** Works better for large vector dimensions.

4 Self-Attention

- **Definition:** Each input element attends to **all elements in the same sequence**.
 - **Use case:** Transformers Encoder (BERT, GPT)
 - **Pros:** Captures **long-range dependencies**; fully parallelizable.
-

5 Cross-Attention

- **Definition:** Query comes from one sequence, Key & Value from another.
 - **Use case:** Transformer Decoder (machine translation)
 - **Pros:** Allows decoder to focus on relevant encoder outputs.
-

6 Multi-Head Attention (MHA)

- **Definition:** Runs multiple attention mechanisms (heads) in parallel.
- **Formula:**

$$MHA(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) W_o$$

- **Use case:** Transformers (BERT, GPT, ViT)
- **Pros:** Captures different types of relationships simultaneously.

7 Sparse / Local Attention

- **Definition:** Only attends to selected or nearby tokens instead of the full sequence.
- **Use case:** Longformer, BigBird
- **Pros:** Reduces computation for **long sequences**.

8 Relative & Positional Attention

- **Definition:** Attention depends on **relative positions** of tokens rather than absolute.
- **Use case:** Transformer-XL, T5
- **Pros:** Better at modeling **sequence order**.

9 Linear / Efficient Attention

- **Definition:** Approximates full attention to reduce $O(n^2) \rightarrow O(n)$ complexity.
- **Use case:** Performer, Linformer
- **Pros:** Handles **very long sequences efficiently**.

💡 Quick Summary Table

Type	Key Idea	Use Case
Additive	Feedforward scoring	Seq2seq (RNN)
Dot-Product	Q·K	NLP Translation
Scaled Dot-Product	$Q \cdot K / \sqrt{d}$	Transformers
Self-Attention	Within same sequence	BERT, GPT
Cross-Attention	Between sequences	Decoder in Transformers
Multi-Head	Multiple parallel attentions	BERT, GPT, ViT
Sparse/Local	Attend to subset	Longformer, BigBird
Relative/Positional	Uses relative position	Transformer-XL, T5
Linear/Efficient	Low-cost attention	Performer, Linformer

🎯 Why Attention is Used in Deep Learning?

1 Focus on Important Parts

- Attention allows the model to **look at the most relevant parts** of input.
- Example (NLP):

Sentence: “The cat sat on the mat because it was tired.”
When predicting “it”, the model **attends** to “cat” instead of all words equally.

2 Handles Long Sequences

- Old RNN/LSTM models forget long-term dependencies.
- Attention **remembers important information from anywhere in the sequence**.
- Example: Translating a long paragraph → attention can find connections between words far apart.

3 Parallel Computation

- Self-Attention in Transformers can process all tokens **at the same time**, unlike RNNs (one by one).
-  Faster training
-  Works well for large datasets

4 Context-Aware Representations

- Each token/word’s representation now **includes information from other relevant tokens**.
- Example:

“Bank” in “river bank” vs “bank account” → meaning changes based on context.

5 Improves Accuracy

- Helps the model make **better predictions** by focusing on key parts.
- Used in:
 - Machine Translation
 - Text Summarization
 - Image Captioning
 - Speech Recognition

6 Flexibility Across Domains

- NLP → text sequences
- CV → image patches (Vision Transformers)
- Audio → time frames
- Multimodal → text + image (CLIP)

💡 In short:

Attention = “**Learn what to focus on**”
Models don’t treat all input equally → **better performance, context understanding, and efficiency**.

🌟 Applications of Attention

Attention is used wherever a model needs to **focus on important information in a sequence or data**.

1 Natural Language Processing (NLP)

Task	How Attention Helps	Examples
Machine Translation	Focus on relevant source words while generating target words	Google Translate, MarianMT
Text Summarization	Attend to important sentences/phrases	BERTSUM, Pegasus
Question Answering	Focus on parts of context relevant to the question	BERT, RoBERTa
Language Modeling	Helps in predicting next word using context	GPT, LLaMA

2 Computer Vision (CV)

Task	How Attention Helps	Examples
Image Classification	Attend to important image patches	Vision Transformers (ViT)
Object Detection	Focus on regions where objects are likely	DETR
Image Captioning	Attend to relevant areas while generating text	Show, Attend and Tell
Segmentation	Highlight important regions for mask prediction	Swin Transformer

3 Speech & Audio Processing

Task	How Attention Helps	Examples
Speech Recognition	Focus on relevant frames in audio	Wav2Vec 2.0
Speech Synthesis	Align text & audio	Tacotron
Speaker Verification	Attend to speaker-specific features	Attention-based models

4 Multimodal AI

Task	How Attention Helps	Examples
Text + Image	Align image regions with text	CLIP, Flamingo
Video + Text	Focus on relevant frames for captions	Video Transformers
Audio + Text	Align speech and transcript	Whisper

5 Other Domains

Domain	How Attention Helps	Examples
Reinforcement Learning	Focus on important states/actions	Decision Transformers
Graph Data	Learn relationships in graphs	Graph Attention Networks (GAT)
Healthcare	Focus on relevant patient records/features	Medical NLP & CV tasks

💡 Why Attention Works Everywhere

- Can handle **variable-length sequences**
- Learns **context-aware representations**
- Supports **parallel computation** → faster training
- Can focus on **key elements** in any type of data

1 Self-Attention from Scratch (Step-by-Step)

python

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```
import torch
import torch.nn.functional as F

# ♦ Input: 3 tokens, each of size 4
x = torch.tensor([
    [1.0, 0.0, 1.0, 0.0], # Token 1
    [0.0, 2.0, 0.0, 2.0], # Token 2
    [1.0, 1.0, 1.0, 1.0]  # Token 3
])

# ♦ Linear weights for Q, K, V (trainable in real model)
W_Q = torch.randn(4, 4)
W_K = torch.randn(4, 4)
W_V = torch.randn(4, 4)

# ♦ Compute Q, K, V
Q = x @ W_Q
K = x @ W_K
V = x @ W_V

# ♦ Step 1: Compute attention scores (dot product)
scores = Q @ K.T

# ♦ Step 2: Scale by sqrt(d_k)
d_k = K.shape[-1]
scores = scores / torch.sqrt(torch.tensor(d_k, dtype=torch.float32))

# ♦ Step 3: Softmax (attention weights)
attn_weights = F.softmax(scores, dim=1)

# ♦ Step 4: Weighted sum of V
output = attn_weights @ V

print("Attention Weights:\n", attn_weights)
print("\nOutput (Context Vectors):\n", output)
```

Explanation

- `Q, K, V` → Queries, Keys, and Values
- `scores` → Measure of how much one token attends to another
- `softmax` → Converts scores to probabilities
- `output` → Weighted sum → context-aware representation

⚙️ 2 Using PyTorch's Built-in MultiHeadAttention

```
python📄 Copy code  
  
import torch  
import torch.nn as nn  
  
# 💡 Define Multi-Head Attention layer  
attention = nn.MultiheadAttention(embed_dim=8, num_heads=2, batch_first=True)  
  
# 💡 Example input: batch_size=1, seq_len=3, embedding_dim=8  
x = torch.randn(1, 3, 8)  
  
# 💡 Self-attention: Q = K = V = x  
output, weights = attention(x, x, x)  
  
print("Attention Weights:\n", weights)  
print("\nOutput:\n", output)
```

🔍 Explanation

- `nn.MultiheadAttention` automatically handles Q, K, V projections.
- `num_heads=2` → multi-head version (captures different relationships).
- Used in Transformers (BERT, GPT, ViT, etc.).

💡 What Happens Internally

```
mathematica  
  
Input → Q, K, V Linear layers  
      ↓  
Q·K^T / √d_k → Softmax → Weights  
      ↓  
Weights·V → Context Vector → Output
```

✅ Output Meaning

- **Attention Weights:** how much each token focuses on others
- **Output:** new representation of each token with contextual info